

↗ https://github.com/irisyy/APpliedML_IBD

APPLIED ML: CLINICAL STUDIES IN PRACTICE

DATABASE OVERVIEW



HMP2 - IBDMDB

[nature](#) > [articles](#) > article

Article | [Open access](#) | Published: 29 May 2019

Multi-omics of the gut microbial ecosystem in inflammatory bowel diseases

[Jason Lloyd-Price](#), [Cesar Arze](#), [Ashwin N. Ananthakrishnan](#), [Melanie Schirmer](#), [Julian Avila-Pacheco](#), [Tiffany W. Poon](#), [Elizabeth Andrews](#), [Nadim J. Ajami](#), [Kevin S. Bonham](#), [Colin J. Brislawn](#), [David Casero](#), [Holly Courtney](#), [Antonio Gonzalez](#), [Thomas G. Graeber](#), [A. Brantley Hall](#), [Kathleen Lake](#), [Carol J. Landers](#), [Himel Mallick](#), [Damian R. Plichta](#), [Mahadev Prasad](#), [Gholamali Rahnavard](#), [Jenny Sauk](#), [Dmitry Shungin](#), [Yoshiki Vázquez-Baeza](#), [IBDMDB Investigators](#), ... [Curtis Huttenhower](#)  [+ Show authors](#)

[Nature](#) **569**, 655–662 (2019) | [Cite this article](#)

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- longitudinal tracking of Crohn's and ulcerative colitis (subtypes of inflammatory bowel disease (IBD))
- multi-omics data: metagenomics, metatranscriptomics, metabolomics, proteomics, viromics, 16S rRNA
- Diet surveys, blood, fecal calprotectin data
- HMP2 metadata: Total 5533 samples, 490 features
- ~ 90% have missing values: Treatment, Classification, Visit-specific fields
- Less than 10% have no missing values
- Split by participant (not by row) ---- Have 131 unique participants

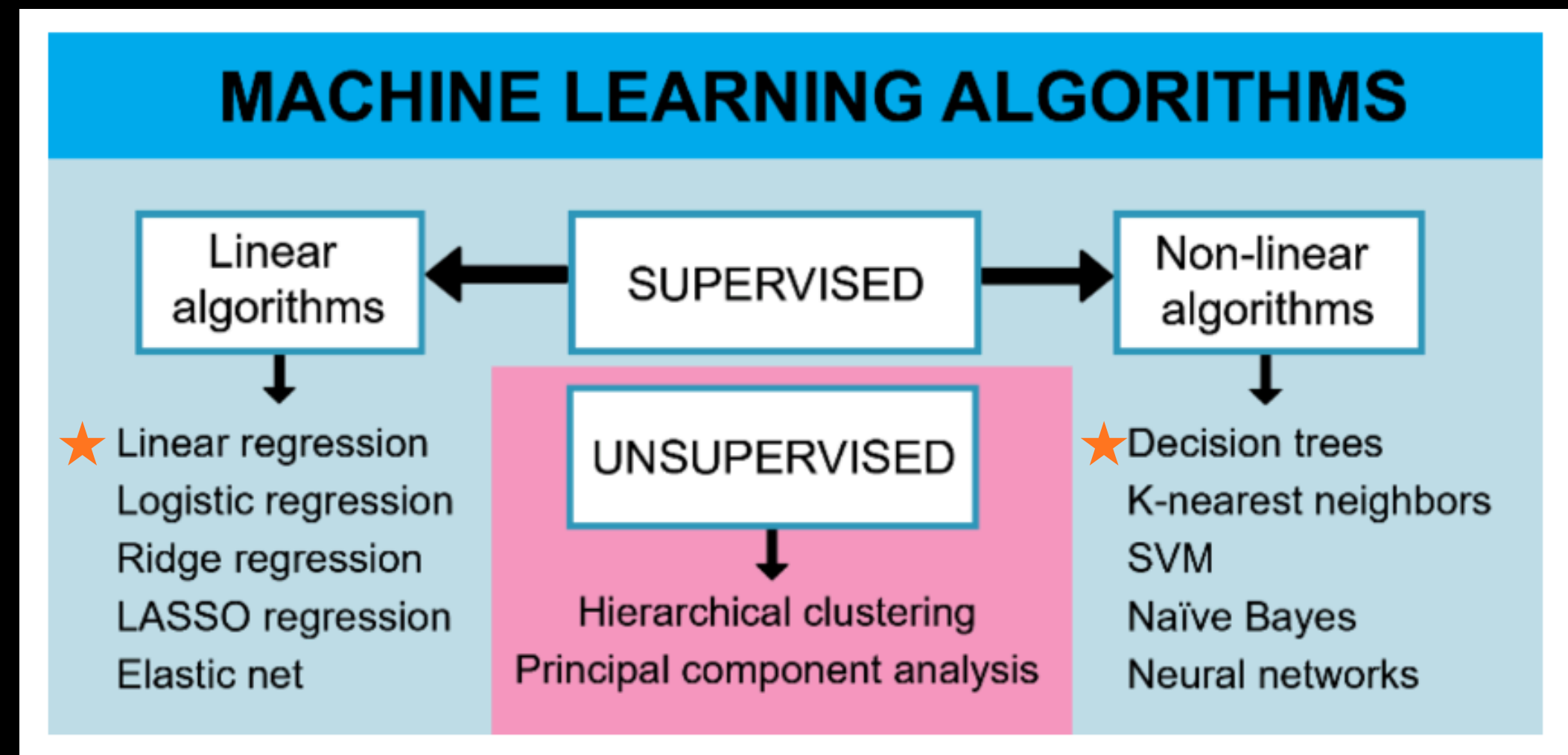
RESEARCH QUESTION

Use metadata and omics data (e.g. metagenomics) to predict diagnosis label (UC, CD, nonIBD)

APPROACH

- Use one layer of omics data
- Work with no missing values data

METHOD



WORKFLOW



DATA PREPROCESSING

- Download
- Parsing
- Species-level filtering
- Subject-level splitting
- one-omics layer
- no missing values

Baseline Model

Model
development

- RandomizedSearchCV
- LightGBM

Model
comparison

- macro F1
- accuracy

BASELINE MODEL



LINEAR REGRESSION

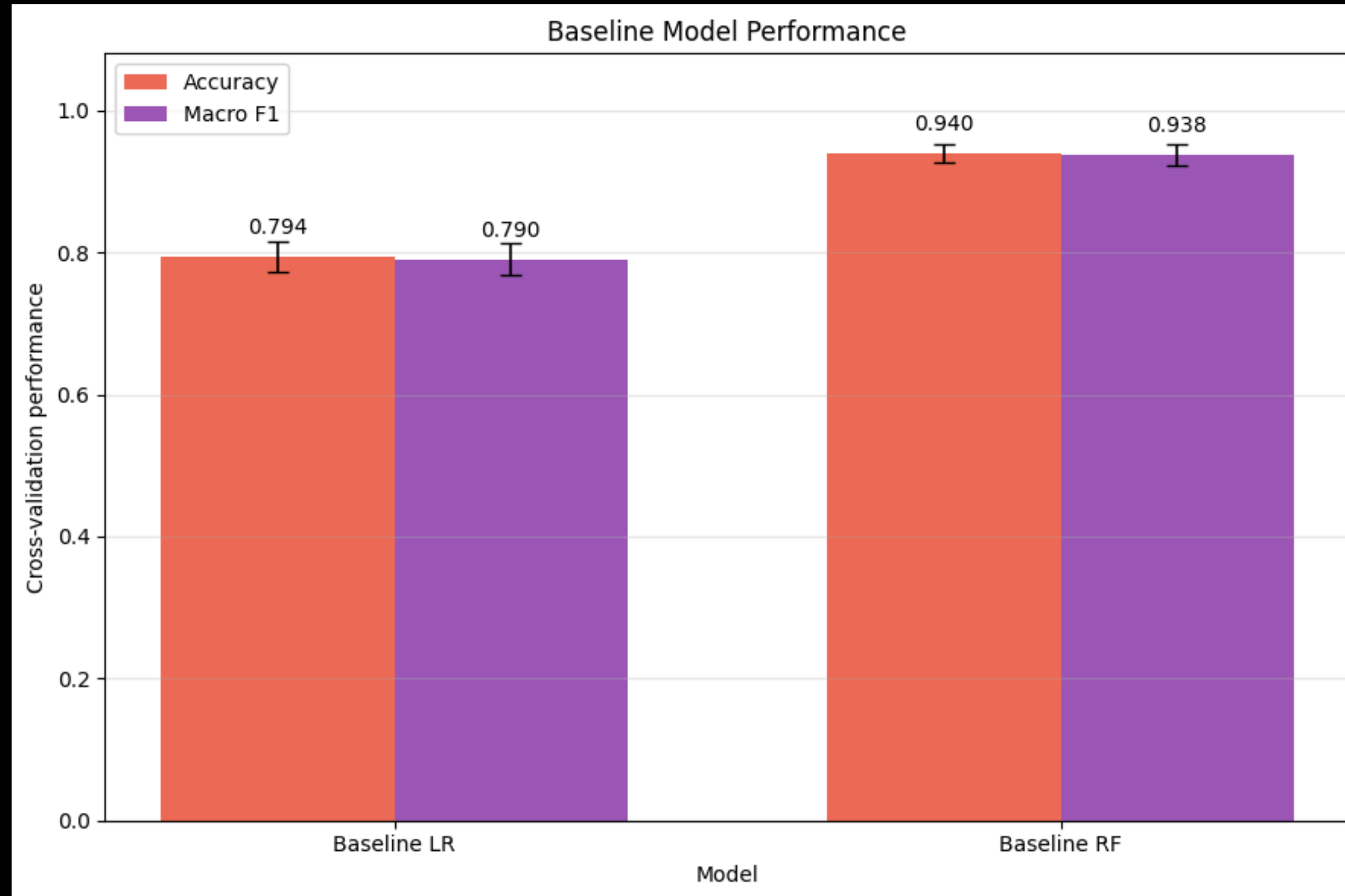
- linear effects
- fast to train and easy to reproduce

RANDOM FOREST

- nonlinear relationship
- works well with high-dimensional taxa data

```
1 # =====
2 # Baseline LR and RF
3 # =====
4 lr_base = Pipeline([
5     ("imputer", SimpleImputer(strategy="median")),
6     ("scaler", StandardScaler(with_mean=False)),
7     ("clf", LogisticRegression(max_iter=3000, class_weight="balanced"))
8 ])
9
10 rf_base = Pipeline([
11     ("imputer", SimpleImputer(strategy="median")),
12     ("clf", RandomForestClassifier(
13         n_estimators=300,
14         random_state=42,
15         class_weight="balanced_subsample",
16         n_jobs=-1
17     ))
18 ])
19
20 lr_cv = evaluate_cv(lr_base, X_train, y_train, groups_train, "Baseline Logistic
Regression")
21 rf_cv = evaluate_cv(rf_base, X_train, y_train, groups_train, "Baseline Random
Forest")
```

BASELINE MODEL



MODEL DEVELOPMENT



HYPERPARAMETER TUNING

RandomizedSearchCV

- lightweight
- reproducible
- integrated directly with the scikit-learn pipelines and group cross-validation scheme

3-fold group cross-validation

n_iter=3

```
1 # =====
2 # Light hyperparameter tuning - RF
3 # =====
4
5 # Random forest tuning
6 rf_pipe = Pipeline([
7     ("imputer", SimpleImputer(strategy="median")),
8     ("clf", RandomForestClassifier(
9         random_state=42,
10        class_weight="balanced_subsample",
11        n_jobs=-1
12    ))
13 ])
14
15 rf_params = {
16     "clf__n_estimators": randint(200, 800),
17     "clf__max_depth": [None, 10, 20, 30, 40],
18     "clf__min_samples_split": randint(2, 20),
19     "clf__min_samples_leaf": randint(1, 10),
20     "clf__max_features": ["sqrt", "log2", 0.3, 0.5, None]
21 }
22
23 rf_search = RandomizedSearchCV(
24     rf_pipe,
25     param_distributions=rf_params,
26     n_iter=3,
27     scoring="f1_macro",
28     cv=gkf,
29     n_jobs=-1,
30     random_state=42,
31     verbose=1
32 )
```

```
1 # =====
2 # Light hyperparameter tuning - LR
3 # =====
4
5 # Logistic regression tuning
6 lr_pipe = Pipeline([
7     ("imputer", SimpleImputer(strategy="median")),
8     ("scaler", StandardScaler(with_mean=False)),
9     ("clf", LogisticRegression(max_iter=5000, class_weight="balanced"))
10 ])
11
12 lr_params = {
13     "clf__C": np.logspace(-3, 2, 20),
14     "clf__penalty": ["l2"],
15     "clf__solver": ["lbfgs", "saga"]
16 }
17
18 lr_search = RandomizedSearchCV(
19     lr_pipe,
20     param_distributions=lr_params,
21     n_iter=3,
22     scoring="f1_macro",
23     cv=gkf,
24     n_jobs=-1,
25     random_state=42,
26     verbose=1
27 )
```

MODEL DEVELOPMENT

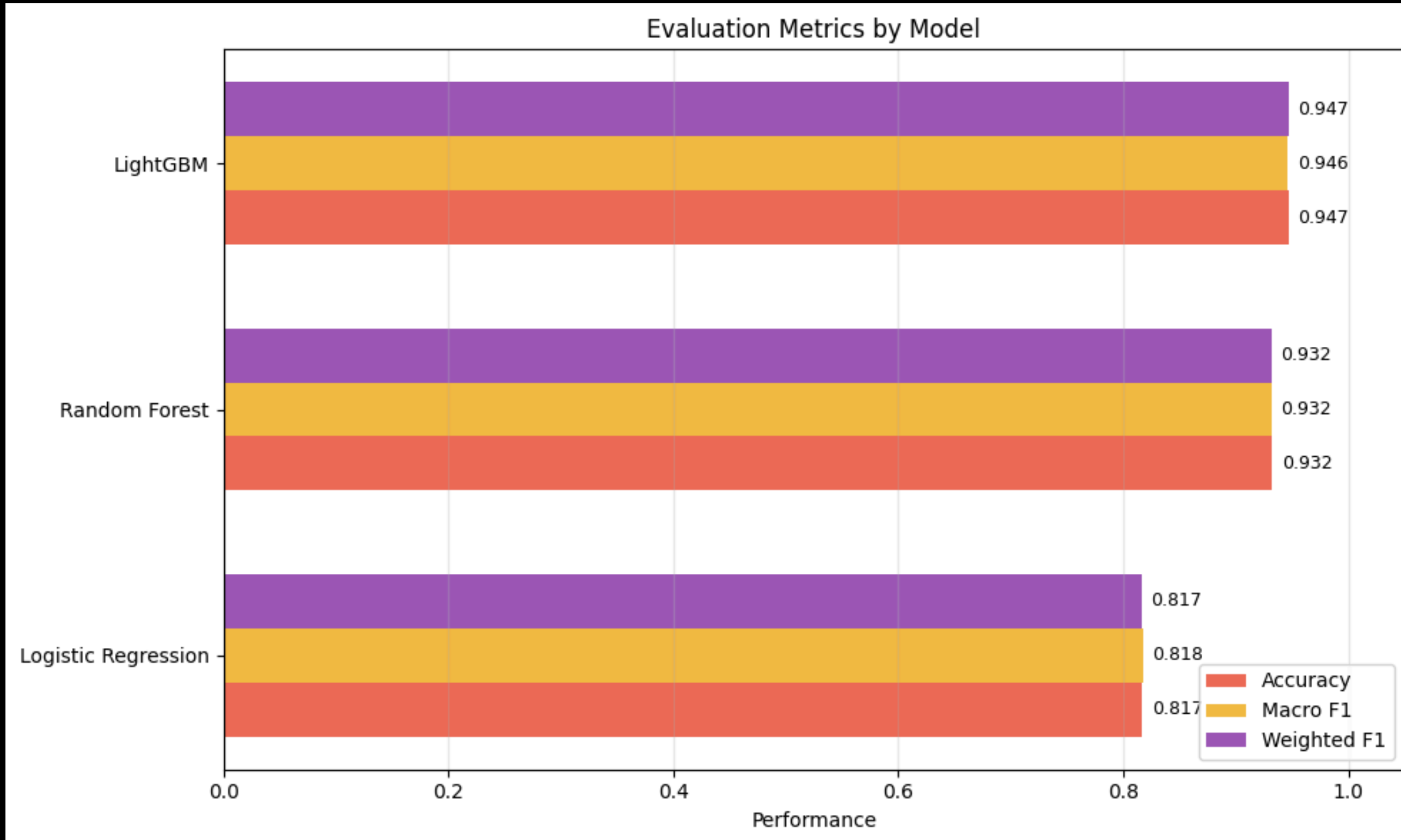


```
6 lgbm = Pipeline([
7     ("imputer", SimpleImputer(strategy="median")),
8     ("clf", lgb.LGBMClassifier(
9         objective="multiclass",
10        num_class=3,
11        random_state=42,
12        n_jobs=-1
13    ))
14 ])
15
16 lgbm_params = {
17     "clf__n_estimators": randint(200, 1000),
18     "clf__learning_rate": uniform(0.01, 0.2),
19     "clf__num_leaves": randint(15, 63),
20     "clf__max_depth": [-1, 5, 10, 15],
21     "clf__min_child_samples": randint(5, 50),
22     "clf__subsample": uniform(0.6, 0.4),
23     "clf__colsample_bytree": uniform(0.6, 0.4)
24 }
25
26 lgbm_search = RandomizedSearchCV(
27     lgbm,
28     param_distributions=lgbm_params,
29     n_iter=3,
30     scoring="f1_macro",
31     cv=gkf,
32     n_jobs=-1,
33     random_state=42,
34     verbose=1
35 )
```

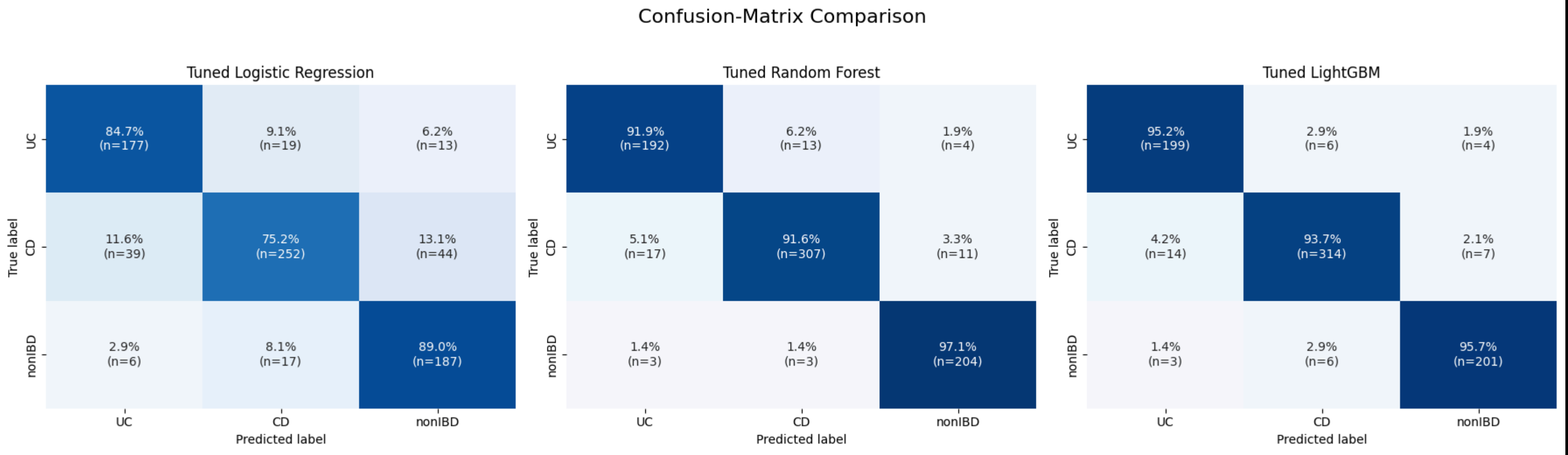
ADVANCED MODEL: LIGHTGBM

- fast
- gradient-boosting model
- efficient tree-based learning
- works well with high-dimensional data and efficient on large-scale data
- effective in tabular classification problems

MODEL PERFORMANCE

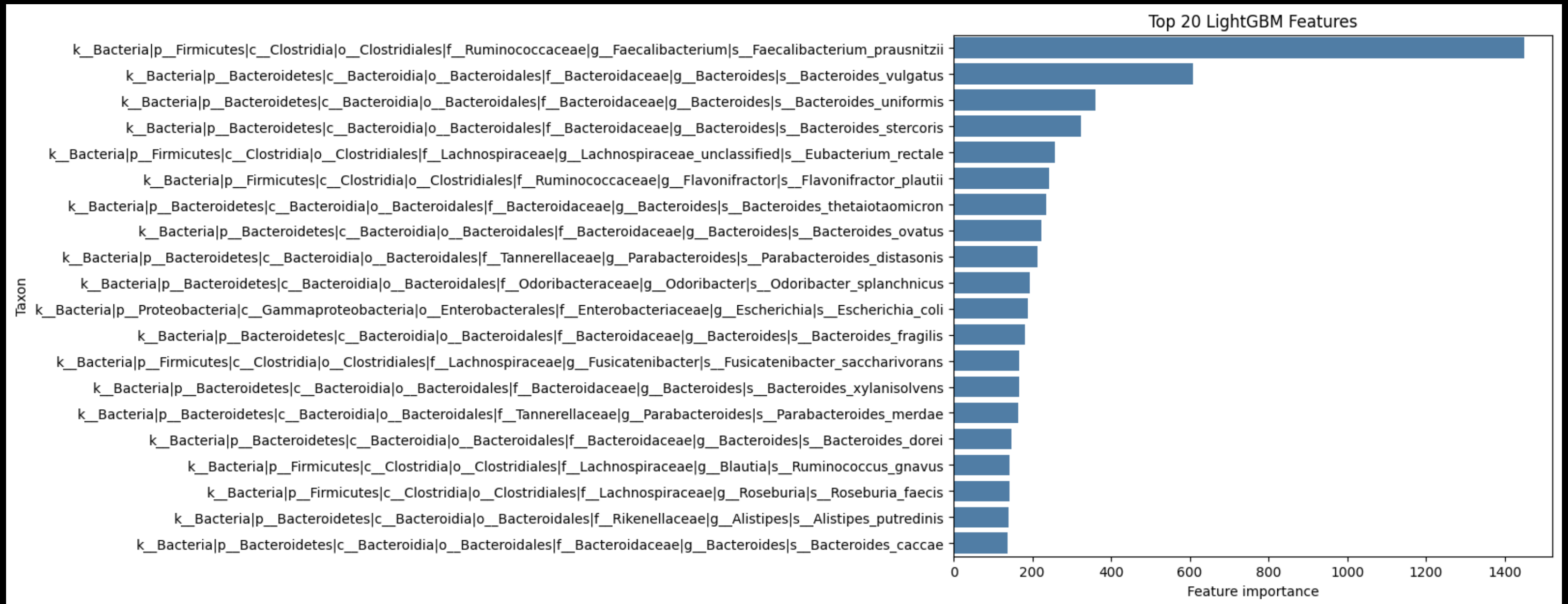


MODEL PERFORMANCE



Best model: LightGBM

MODEL PERFORMANCE



Most informative taxa: *Bacteroides* species, *Faecalibacterium prausnitzii*, *Akkermansia muciniphila*, *Roseburia* spp., and *Ruminococcus/Lachnospiraceae* members

CHALLENGES



**MISSING
VALUES IN
LONGITUDINAL
STUDIES**

**LABEL
LEAKAGE**

**HYPERPARAMETER
TUNING &
OVERFITTING**

**SUBJECT-
LEVEL SPLIT**

**LIMITED
CLINICAL
ENDPOINT
INFORMATION**

**CLASS
IMBALANCE**

KEY TAKEAWAYS

- 01** Group split by participant ID → ensures all samples from the same subject are in either the train or the test set, preventing data leakage.
- 02** High quality, clean and relevant input data to achieve best model performance – garbage in, garbage out.
- 03** Tree-based models (random forests and LightGBM) are well-suited for handling the high dimensionality and nonlinearity of microbiome data.



LIMITATION

Single omics only (metagenomics)

Small-to-moderate sample size relative to feature count → Potential overfitting and leakage risk

No handling of missingness and sparsity

FUTURE PROJECTS

Integration of more omics layer

External validation on independent IBD metagenomic datasets

Longitudinal modeling: Apply time-aware models to capture temporal dynamics

Thank you for your attention!